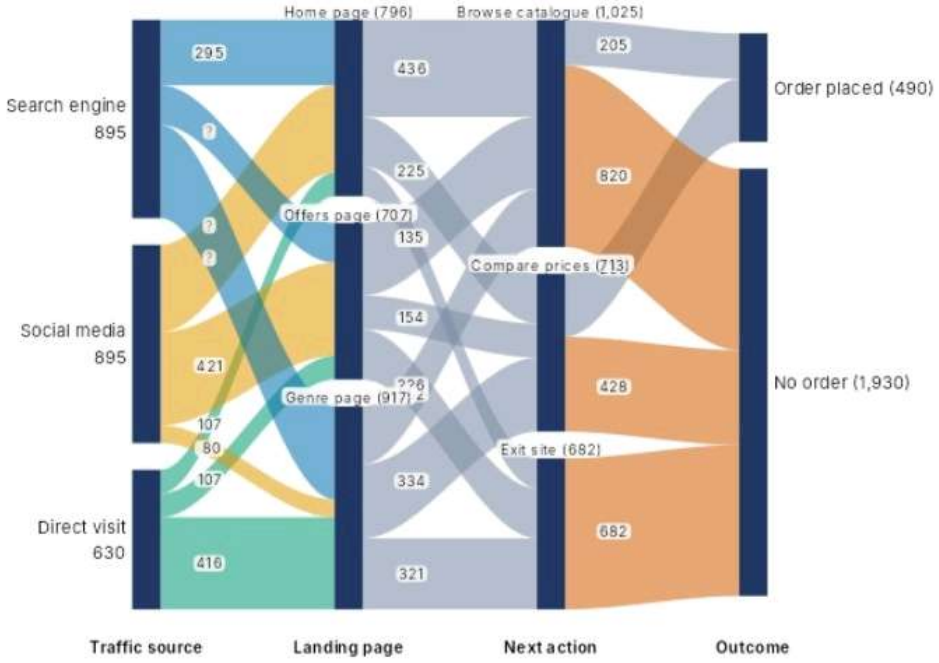
DSA06 - Data Handling and Visualization | Activity 6: Website Flow Sankey Design | Slot C

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Scenario: The analytics team of an online bookstore tracked 2,420 sessions last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of sessions on every flow. Three flows are hidden and marked "?".



(a) Using conservation of flow (what enters a node equals what leaves it) find the three hidden flows, showing working: Search engine to Offers page; Search engine to Genre page; Social media to Home page. State the rule you used. [4]

(b) Compute: (i) the overall conversion rate, i.e. Order placed as a percentage of all sessions; (ii) the largest single flow into "No order" or "Exit site"; (iii) the percentage of "Offers page" sessions that go on to "Browse catalogue". [4]

(c) Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data. [4]

(d) The Sankey is to be printed with 1 cm of node height = 100 sessions. Compute the height in cm of each landing-page node (Offers page, Home page, Genre page). [2]

(e) Write R code: a links data frame for the first four flows leaving "Search engine" and "Social media" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call (or plotly) that draws the whole diagram. [4]

(f) State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each. [2]

Activity - 6

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(a) Find 3 Hidden Flows

1. Search engine -> offers page = ?
   - Total Search engine sessions = 895
   - Search engine -> Home page = 245
   - Search Engine -> Genre page = ? (hidden)
   - Search engine -> offers page = ?
2. Search engine -> Genre page = ?
3. Social Media -> Home page = ?

(b) Calculations

(i) Conversion Rate: $\frac{490}{2420} \times 100$

$= 20.25\%$

(ii)
- No order: 1,930 sessions
- Exit site: 682 sessions

(iii) Percentage of offers page sessions that go to "Browse Catalogue".

$\frac{436}{707} \times 100 = \boxed{61.67\%}$

(C) Four Design Decisions

1. Node Order: Arrange traffic source, landing page, next action, and outcome from left to right to show the user journey.
2. Colour Rule: Use different colours for traffic sources and orange for unsuccessful outcomes
3. Labels: Display node names and session count clearly.
4. Highlight: Highlight the 440 orders and 1,930 no-order sessions to compare conversation and drop-off.

(D) Node Heights

1 cm = 100 session.

$$\text{Height} = \frac{\text{Sessions}}{100}$$

- Offers page = 707/100 = 7.07cm
- Home page = 796/100 = 7.96cm
- Genre page = 917/100 = 9.17cm.

(E) R Code

```
library (networkD3)
links <- data.frame(
  source = c(0, 0, 1, 1), target = c(3, 4, 3, 4),
  value = c(295, 200, 421, 436)
)
nodes <- data.frame (name = c ("search", "Email", "Direct",
  "Home", "Offers"))
)
SankeyNetwork (Links: links, Nodes: nodes,
```

Source = "source", Target = "target",
Value = "value", NodeID = "name")

(f)

1. Too many flows: The diagram becomes cluttered and difficult to read.
   Alternative: Bar chart.

2. Very small flows: Small flows may be difficult to identify. Alternative: Table.